Modeling Earth Systems and Environment (2022) 8:1419–1431
https://doi.org/10.1007/s40808-021-01150-3

ORIGINAL ARTICLE

Modelling urban expansion of a south-east Asian city, India: comparison between SLEUTH and a hybrid CA model

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Received: 28 November 2020 / Accepted: 15 March 2021 / Published online: 4 May 2021

Abstract

Accurate identification of the dominant driving factors in the expansion of a city is essential for cellular automata (CA)-based urban expansion modelling. The current study aimed at analyzing the drivers of change for Vijayawada city, a tier-II developing city in south-east Asia, India and then applies them to model the future scenario of the city in a CA-based environment. Furthermore, the study also aims at comparing the efficiency of the renowned CA-based urban growth model, SLEUTH (Slope, Land use, Exclusion, Urban extent, Transportation, Hill shade), and a self-designed CA-based hybrid model developed in combination with genetic algorithm (GA) for developing future scenario of the city. The models were built, calibrated, and validated for the city using land use maps derived from temporal Landsat satellite images. Analysis of the statistical significance of the driving factors shows that the terrain and population density are the two dominant factors influencing expansion of the city. Both the models output on predicting the urban growth indicate that the available open spaces within the existing city extent gets further converted to built-up indicating infill development, and more growth occurs along the fringes of the existing city. Also, emergence of new urban centers merged with the existing city areas for predictions till 2050 and extended into the dense vegetation on the north resulting in further reduction of green cover around the city. Validation of both the models showed overall accuracy higher than 85%. The inclusion of the demographic variable and utilization of the GA ensured that the hybrid model performs better than the SLEUTH model. Analysis on the built-up densification of the cityscape showed a shift in the pattern of built-up dynamics over time. The built urban expansion scenario for Vijayawada informs urban modelers and policymakers the possible evolution of the city.

Keywords Vijayawada · Urban expansion · SLEUTH model · Cellular automata model · Urban footprint

Background

Urbanization is a form of metropolitan development causing irretrievable changes to the physical geography of an area as a consequence of perplexing economic, social and political forces (Zadbagher et al. 2018). Growth of an urban area indicates a transformation of the natural environment including the agricultural and open areas into built-up areas (Bhatta 2009). This development especially along the existing urban edges causing expansion and sprawl of the city serves as a major environmental challenge in most countries (Dadashpoor and Salarian 2018). Several studies have shown negative consequences of urban expansion like loss of naturally vegetated surfaces, agricultural and environmentally sensitive areas (Chen et al. 2003; Johnson 2001), reduction of environmental quality with increased air and noise pollution (Raje et al. 2018), and rise in land surface temperature (Vani and Prasad 2020). Though sprawl in general, is related with the peri-urban areas of large cities, expansion of utilities beyond the existing city extent at the cost of natural resources also pose similar scenario (Cadieux and Taylor 2013; Adamiak 2016; Bose and Chowdhury 2020).

LUCC (land use-cover changes) are dynamic spatio-temporal processes involving multiple factors that influence the changes at different spatial extents. These factors may be termed as the drivers or the driving forces of land use (LU) change and may be categorized broadly as proximate—that have a direct influence on the land use-land cover (LULC) systems and cause changes; and underlying—that have indirect impacts on the LULC processes. These various groups

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of drivers were found to interact directly via feedbacks, and thus often have spatio-temporal synergetic effects (Lambin et al. 2003; Steffen et al. 2004).

Predicting the trend of land use transition is a key consideration in the field of urban planning (Osaragi and Kurisaki 2000). Hence, models are being used to analyse the historic and future trends and consequences of LUCC as a simplified representation of the real world (Benders 1996). Besides, models help to interpret the process of urbanization and provide information on the future state of cities under different scenarios (Lambin et al. 2006; Koomen et al. 2007; Noszczyk 2018). Urban expansion modelling exposes the underlying processes that cause spatio-temporal changes (Akin et al. 2015; Liu et al. 2017), providing planners and decision-makers with early warning of ecological and environmental consequences (Sun et al. 2018).

There exist different modelling approaches. Choice of the suitable model depends on the objective of the analysis, availability of the data and resources. Optimization models, the most suitable models for policy decisions, develop an optimum solution for a given system and are based on the assumption that the agents of the system optimize. Though these models better replicate the temporal dynamics of the system, they pose difficulties in representing the spatial aspects (Noszczyk 2018). Markov models, the most accurate models to quantify LUCC between two periods does not account for the spatial distribution of the LUCC (Muller and Middleton 1994; Lantman et al. 2011; Iacono et al. 2015; Louca et al. 2015; Dang and Kawasaki 2016). Agent-Based Models (ABM)/Multi-agent Simulation (MAS) are exclusively involved in studies on complex processes to define the relationship between agents (human decision-making entities) and their environment. ABMs emphasis on the clearest and easily quantifiable aspects of LU and ignores factors such as migration, the use of input data, and the impact of regional and global economic variables (Dang and Kawasaki 2016).

Cellular Automaton (CA) models are simple two-dimensional discrete models and are better in replicating the real-world spatial patterns and local spatial processes (Lagarias 2012; Qiang and Lam 2015). A spatially defined CA model is appropriate for looking into the dynamics of urban growth (Noszczyk 2018). The SLEUTH (Slope Landuse Exclusion Urban Transportation Hillshade), a CA-based model is a universally applicable model to study the evolution of cities (Silva and Clarke 2002) and is globally adopted for urban growth modelling (Qi 2012; Kong et al. 2019; Chandan et al. 2019). The transition rules for the CA model are built based on the driving forces of LUCC. Factor selection in building the model is vital as the inclusion of too many factors in the model leads to multi-collinearity effect thus degrading the efficiency of the model (Feng and Tong 2017; Thaden et al. 2018; Feng et al. 2019). To overcome this issue, appropriate selection of the dominant driver of the LUCC is essential. This helps not only in reducing the variable multi-collinearity effects on model results but also minimizes information loss.

Though the CA algorithm is an effective tool to reconstruct and model future scenarios of urban expansion by analysing the LUCC (Soares-Filho et al. 2013; Liu et al. 2018), the robustness of the CA algorithm used autonomously to produce convincing results on built-up dynamics has degraded over time. This paved the way to the emergence of hybrid models of CA. Hybrid models include different modelling approaches fitting into a single modelling framework to apply multiple concepts of different approaches (Saxena et al. 2018; Saxena and Jat 2018).

The rationale and objectives of the current study

The south-east Asian city, Vijayawada of the state, Andhra Pradesh (AP), India is a hub for educational institutions and witnesses' rapid urbanization and population migration (Vani and Prasad 2020). Over time, the city has agglomerated at the expense of natural vegetation, hills, and water bodies leading to loss of green spaces and wetlands that existed within the city. Sante et al. (2010) and Aburas et al. (2016) on reviewing CA urbanization models conclude that the most suitable way to study urban dynamics pattern is through the utilization of the CA approach. Towards this, the objectives of the current study are to (1) identify the dominant drivers of urban expansion for Vijayawada, (2) model an appropriate scenario of the future of the city, Vijayawada using an efficient discrete-time CA-based dynamic spatial model, (3) identify the pattern of city growth, (4) quantify the built-up dynamics with respect to the city growth, and (5) compare the efficiency of the renowned urban growth model—SLEUTH—and a self-designed hybrid urbanization model based on a CA—Genetic algorithm (GA)—multinomial logistic regression (MLR) environment.

Study area

The study area (Fig. 1) Vijayawada (16° 31′ 9.48″ N longitude, 80° 37′ 49.8″ E latitude), presently the commercial headquarter of the AP state is located at an elevation of 11 m above the mean sea level on the banks of River Krishna. The city houses hills and canals that emerge from the Prakasam barrage built across the Krishna River: Eluru, Bandar, and Ryves that run through the heart of the city. The city experiences extreme temperatures during summer (40 °C and higher) and winter (18 °C) (IMD 2015). The Government of India (GoI) ranks Indian cities through a classification system prescribed by the sixth central pay commission that categorizes cities into Tier-I, Tier-II, Tier-III, and Tier-IV. The cities were classified based on their population reported by the census 2011. Concerning the census estimates, Vijayawada city was upgraded to Tier-II category of cities in 2014. Vani

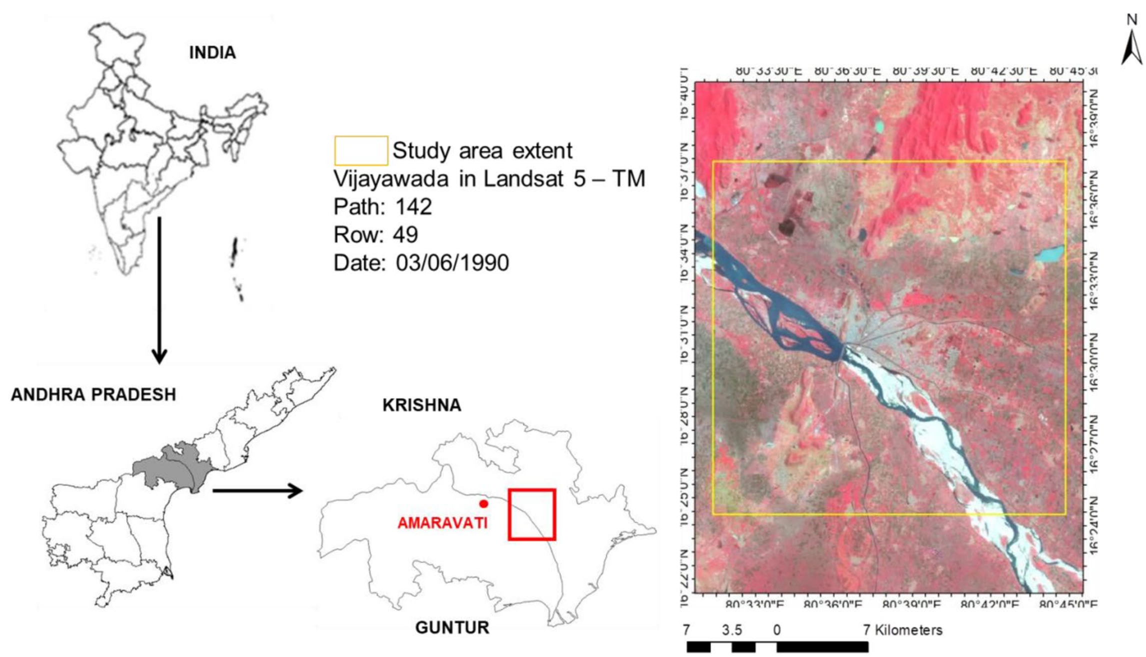

Fig. 1 Study area

and Prasad (2020) reported that the built-up area of the city has increased from 28.20 km^2 in 1990 to 138.01 km^2 in 2018 at the cost of natural vegetation in and around the city extent. This finding is in agreement with the recordings of Asad and Mehrotra (2016) that states that a significant proportion of future urban growth would occur in areas defined other than the Tier-I cities. This poses a serious concern on the future state of the Tier-II city, Vijayawada for the built-up growth.

Methods

The LULC maps derived from Landsat satellite imagery viz., 1990, 2000, 2010, and 2018 (further details refer Vani and Prasad 2020) has been used as an input to train and validate the models to predict the scenarios of LULC in 2020 and 2030 under the influence of bio-physical, demographic, socio-economic, and secondary drivers of LUCC. The intervals 1990–2000 and 2000–2010 were used for calibrating and validating the model, respectively. The bio-physical drivers are the ecological forces that influence changes in LULC either directly or indirectly. Secondary drivers are the factors derived from the bio-physical and socio-economic drivers of LUCC. In the current study, slope, elevation, and hill shade are used as bio-physical drivers; population density and richness index based on the per capita income (PCI) as socio-economic drivers; and distance to road and city center (as observed in 1990 built-up extent) as secondary drivers of urban expansion. Traditionally, CA models work with raster datasets. Hence, all the datasets utilized in this study are in raster formats. Furthermore, all the input data (Table 1) were harmonized to the same spatial data format (raster), resolution (30 m × 30 m), projection (Universal Transverse Mercator), and extent.

Table 1 Input parameters used in the model

Data	Source	Path/Row	Date of acquisition
LULC 1990	Landsat TM	142/49	3rd June
LULC 2000	Landsat TM	142/49	27th April
LULC 2010	Landsat TM	142/49	9th May
LULC 2018	Landsat OLI	142/49	31st May
Bio-physical driver	ASTER DEM		
Socio-economic drivers	Census—1991, 2000 and 2011, World Bank and AP economic development board		
Secondary driver	Open street map		

Urban growth model (UGM)—SLEUTH

SLEUTH is a modified CA urban growth model that effectively predicts the growth of a city under different types and controlling factors viz., Spontaneous growth, New spreading center, Edge growth and Road influenced growth controlled by Dispersion, Breed, Spread, Road gravity coefficients, and Slope of the terrain, respectively (Table 2). As the name of the model indicates, slope of the terrain, hill shade representing the morphology of the study area, historical land use maps (1990, 2000, 2010, and 2018), historical urban extent maps (1990, 2000, 2010, and 2018), exclusion layer (Krishna river) indicating areas restricted for urbanization, and road network map (1990, 2018) satisfy the input data requirements to the model.

Working of the model can be summarized into three phases as Testing phase, Calibration phase and Prediction phase (Clarke et al. 1997). Testing phase checks for the accuracy of the input data requirements; calibration phase involves training and improving the model to achieve the best fit; and the final prediction phase. Prediction mode uses the parameters derived from calibration mode. The forecast of urban growth is a probabilistic map of potential urbanized nonurban grid based on the assumption that growth trend observed in the past would repeat in the future. Predict mode performs a single run using the best fit values over the set number of Monte Carlo iterations. The mean coefficient results of all Monte Carlo iterations are used to forecast the urban growth.

Each parameter set tries to simulate and replicate the historical urban growth by assuming the initial urban extent layer (1990) as a seed layer, and with the subsequent input urban extents (2000, 2010, and 2018) as control points. The performance of the model has been evaluated in different aspects; (1) urban growth: urban area calculated in sq. km., (2) urban growth pattern: compare statistic (ratio of the modelled population of urban pixels in the final year to the actual number of urban pixels for the same year), urban population (number of urban pixels), edges (urban perimeter), cluster (urban cluster edge pixels), mean cluster size and cluster radius (Xmean and Ymean) (Dietzel and Clarke 2007), (3) calibration time: computational efficiency of calibration process in hours, (4) spatial distribution of urban growth, (5) urban pattern index: Lee–Sallee statistics (Lee and Sallee 1970), a reliable shape measure to identify the pattern of urban growth (Saxena and Jat 2018), (6) best-fit growth coefficients: diffusion, breed, spread, slope resistant and road gravity, (7) best fitness measure: OSM (Optimal SLEUTH Metrics), which is a product of ‘compare’, ‘population’, ‘edges’, ‘clusters’, ‘slope’, ‘X-mean’, and ‘Y-mean’ as a measure of best-fit (Dietzel and Clarke 2007), (8) Kappa statistics: overall accuracy of the model outcomes on the actual and modelled urban growth, respectively. The above-mentioned metrics represent a value range of 0–1 to determine the goodness of fit between the simulated growth and the actual growth for the control years. Greater the metric value towards 1 indicates better model calibration.

Hybrid CA-GA-MLR model

Comprehensive analysis of the driving forces that accelerate urban development is largely useful for CA modelling. Logistic regression or logit models are a common approach to model urban expansion (Mustafa et al. 2017). They predict the presence of built-up expansion using a set of quantitative and/or qualitative variables termed as predictors.

CA-based model is the popular approach to quantify the neighbourhood interactions on a dynamic basis, in which each simulation step updates the neighbourhood state (Clarke and Gaydos 1998). However, pure CA models lack in interpreting the dominant driver of change in the process as they focus mainly on explicit quantification of the transitions by taking into account only the immediate neighbours of each cell. This limitation of the CA model can be overcome by integrating the CA with other modelling methods to incorporate both the static drivers and the dynamic neighbourhood interactions (Poelmans and Van Rompaey 2010). Towards this, in the current study, MLR analysis was performed to examine the ability of each driving factor to explain LUCC and rank the factors by their significance. GA has then been used to record the land

Table 2 Summary of growth types and coefficients in SLEUTH model. Adapted from: Jantz et al. (2004)

Growth cycle	Growth type	Growth coefficient	Description
1	Spontaneous	Dispersion	Selection of a random cell as an urban centre
2	New spreading centre	Breed	Generation of two neighbourhood cells for each selected urban centre from spontaneous growth
3	Edge (organic)	Spread	Generation of at least three neighbourhood cells (that pass the spread coefficient and slope resistance tests) along the existing urban centres
4	Road influenced	Road gravity, dispersion, breed	For each selected cell, generation of at least three urban cells that falls within the set road search radius controlled by the road gravity coefficient
Throughout all cycles	Slope	Slope resistance	Effect of slope on reducing the probability of urbanization
Throughout all cycles	Excluded layer	User defined	User defined areas excluded from getting urbanized

transition rules of the CA model. The evolution of the landscape was then analysed for the built-up density of the city.

The road network data of 2018 were extracted from the open street map. City center (as observed in 1990), slope of the terrain extracted from the digital elevation model (DEM) given by the advanced spaceborne thermal emission and reflection (ASTER) at 30 m spatial resolution, richness index derived from the national statistics and AP economic development board, population records from the census 1991, 2001, and 2011 were used as the independent variables that stimulates urban expansion. The richness index was calculated as the average PCI for each ward divided by the average PCI of AP. The PCI measures the average income earned per person in a given area (city, region, country, etc.) in a specified year. The independent variables also include the distance-based factors of change viz., distance to the city center, and road network derived using the euclidean distance tool in ArcGIS. The Krishna River that forms the southern boundary of the city was maintained as a constraint for built-up expansion. The description of the independent or the explanatory factors that influence the LUCC is given in Table 3.

Construction of transition rules for the CA engine

The developed CA-GA is an enhancement of a typical CA by automatic calibration using GA. The current cell state, contiguity cell effects, constraints, and land transition probability collaboratively determine the land transformation in CA. The CA-GA transition rules can be expressed as:

$$\mathrm{Cell}_{ij}^{t+1} = \mathrm{Transf}\left(\mathrm{Cell}_{ij}^{t}, P_{n_{ij}}, \mathrm{Const}, P_{t_{ij}}\right)$$

where, Cell_{ij}^{t+1} is the state of cell ij at time t + 1, Transf is the transition rules function, Cell_{ij}^{t} is the state of cell *ij* at time *t*, $P_{n_{ij}}$ is the effects of neighboring urban cells estimated at Moore 5 × 5 window, Const is the prohibited areas or constraints for development, $P_{t_{ij}}$ is the transition probability.

Table 3 Description of the factors of LUCC used in the hybrid model

Factor	Name	Type	Unit
X_1	Slope	Continuous	Percent rise
X_2	Distance to road	Continuous	Meter
X_3	Distance to city center	Continuous	Meter
X_4	Population density	Continuous	per km^2
X_5	Richness index	Continuous	Percent

Quantification of the transition probability

The temporally stationary and spatially non-stationary characteristic controlled by the driving factors and that indicate the probability of transition as determined by the MLR is given by,

$$P_{t_{ij}} = \sqrt{P_{fij} \times \left(P_{nij}\right)^{\sigma}}$$

where, P_{fij} is the the probability of expansion influenced by the factors of change, σ is the the relative importance of the neighborhood effect.

The change rate per time step was set as the total quantity of new urban cells allocated equally over the number of time steps as there exists no significant difference between equal and different change rates (Poelmans and Van Rompaey, 2010; Mustafa et al. 2014, 2017).

Calibration of the driving forces of expansion

The influence of the factors in converting a non-built-up class into built-up class thereby accelerating LUCC can be determined using MLR. The general form of an MLR is given by,

$$\log_{(k_n)} = \alpha_{k_n} + \beta_{k_n 1}X_1 + \beta_{k_n 2}X_2 + \ldots + \beta_{k_n n}X_n$$

where, $\log\left(k_n\right)$ is the natural logarithm of class k_n versus the reference class $k_\circ$, X_n is the explanatory variables, α_{k_n} is the intercept for class k_n versus the reference class $k_\circ$, β_{k_n} is the regression coefficients.

The independent or the explanatory variables X_n are the factors of LUCC. The MLR analysis yields coefficients (β_{k_n}) that give the relative contribution of each driving force X_n and are interpreted as weights in the formula that generates the probability of conversion for each class as,

$$(P_{fij} \forall Y = k_n) = \frac{\exp\left(\log\left(k_n\right)\right)}{1 + \exp\left(\log\left(k_1\right)\right) + \exp\left(\log\left(k_2\right)\right) + \ldots + \exp\left(\log\left(k_n\right)\right)}$$

where, $(P_{fij} \forall Y = k_n)$ – probability of transition from the reference class to class k_n in the cell ij.

The goodness-of-fit, in terms of quantifying the variations shown by each independent variable (factor) on the LUCC process, was evaluated using the McFadden pseudo-R-square method (Clark and Hosking 1986; Braimoh and Onishi 2007; Lin et al. 2014; Mustafa et al. 2014; Shu et al. 2014).

Quantification of the neighbourhood effects

The most common explicit way to calibrate the neighbourhood effects dynamically by incorporating path dependencies and self-organized developments that are observed dominantly in urban areas can be done using the CA approach. The CA modeling involves Moore neighbourhood that considers 3 × 3, 5 × 5, or more squared neighbours depending on the cell resolution of the study area. This study adopted 5×5 windows to capture the neighbourhood interactions. Quantification of the effects of the neighbouring urban cell was adapted from White and Engelen (2000) as,

$$P_{nij} = \sum_k \sum_x \sum_d w_{kxd} \times I_{kxd}$$

where, w_{kxd} is the weighting parameter assigned to a cell with class k at position x and at distance d to represent the interaction with the urban cell

$$I_{kxd} = \left\{ \begin{array}{c} 1, \quad \text{if the cell in distance d is occupied by class k} \\ 0, \quad \text{otherwise} \end{array} \right\}$$

GA has been utilized to automatically calibrate the neighbourhood weighting parameters (w_{kxd}). Population for calibrating the proposed model includes the number of built-up, vegetation and other class pixels, and distance to the road network in a defined square neighbourhood. CA model was then executed on each member in the population for testing its fitness to satisfy the objective function.

The solution is thus a set of optimal parameters for framing the CA transition rules. These parameters, also termed as Pareto front optimized solutions, are non-dominated to each other but are significantly better than the rest of solutions. Moreover, GA does not make assumptions on the underlying processes. Hence, GA being a heuristic search algorithm, it has been used to auto-calibrate the CA model.

The final urban pattern was modeled based on the total land available to be urbanized at the most recent year (2010 for predicting 2020). The threshold is achieved only if both the maximum number of iterations and the total land availability are achieved.

The transition potential matrices were calibrated for 1990–2000, and the calibration results were then used to simulate the 2000 and 2010 built-up pattern. The simulation results of 2000 and 2010 were compared with the actual 2000 and 2010 maps to validate the allocation accuracy of the calibrated model.

Validation

The residuals are represented by the root mean square error (RMSE) measured between the observed land transition value and the predicted transition probability. The RMSE-based objective function adapted from Feng and Tong (2018) is given as

$$\min f\left(f_1 \ldots f_n\right) = \sqrt[2]{\frac{\sum \left(P_{tp}\left(f_1 \ldots f_n\right) - P_{to}\right)_{ij}^2}{N}}$$

where, $\min f\left(f_1 \ldots f_n\right)$ is the objective function, $P_{tp}\left(f_1 \ldots f_n\right)$ is the predicted transition probability for cell ij, P_{to} is the observed land transition for cell ij, N is the total number of samples.

Data preparation for MLR

The disparity in units, spatial autocorrelation, and multi-collinearity are the three factors that bias the results of the MLR analysis (Mustafa et al. 2015). This necessitates the check for any of their presence in the independent variables before performing regression. Standardizing all the continuous independent variables ensures the estimated weights update similarly rather than at different rates during the process. This eliminates disparity in the units and the scale of X_n.

Spatial autocorrelation issue was overcome by a data sampling approach as suggested by Cammerer et al. (2013), Li et al. (2013). Variance inflation factors (VIF) have been utilized to overcome the effects of multi-collinearity.

Cityscape analysis

Cityscape analysis investigates the urban extent of the city over time from 1990 to 2030 and identifies the pattern changes observed in the built-up dynamics over time. The analysis is primarily to identify the dominant locations of change and the likelihood of development pattern in the future. The LULC maps at an equal interval of 10 years from 1990 to 2030 were reclassified into built-up area, water, and open spaces. The analysis focuses mainly on two phases viz., (1) analyse the spread of the built-up class over time in terms of the urbanness observed, and (2) analysis on the existing open spaces accounting towards urban footprint.

Urbanness is measured as a percentage of built-up class within a radius of 1 km^2 centered on the test pixel. The built-up pixels were further classified based on the observed urbanness as urban built-up (> 50% urbanness), sub-urban built-up (10–50% urbanness), and rural built-up (< 10% urbanness). The pixels other than built-up class and water were categorized into open land class. The open class was further classified into urbanized open land (undeveloped open land with urbanness > 50%), fringe open land (undeveloped patches within 100 m of built-up patches), captured open land (undeveloped patches less than 2 km^2 and surrounded by urban built-up and fringe open land), and rural open land (undeveloped and those patches that are neither fringe nor captured). The urban footprint thus indicates the pervious and open spaces likely to be impacted because of their built-up surroundings.

Results

Scenario of the future city by the SLEUTH—UGM

The model was calibrated with 100 m × 100 m spatial resolution input data with 10 numbers of Monte-Carlo iterations for each calibration phase. The CA operation was performed over 5 × 5 contiguity. The coarse calibration that completed in 22,944.99996 s was for 0–100 coefficient space with a step of

25 for each growth coefficient. The best OSM and Lee-Sallee metric values were found to be 0.025 and 0.350, respectively. The successive growth coefficient values that regulate the model for better calibration is presented in Table 4 below.

The model was trained using the 1990 and 2000 datasets and tested for predictions in 2010 and 2018, and the overall model accuracy achieved was 83.7%. Predictions were done for the years 2020, and 2030 using the best-fit coefficients achieved during model calibration (Fig. 2). Since the SLEUTH model operates in a single time step, the growth of the city per annum until the prediction year (2020/2030) was represented by the model (Fig. 3).

Scenario of the future city by the hybrid CA-GA-MLR model

Identification of the dominant driver of built-up expansion

The MLR model has been applied to investigate the contribution of the factors (independent variables, X) on the probability of changes among the different LULC classes (dependent variables, Y). The factors considered potential in accelerating LUCC were standardized and tested for multi-collinearity effects before being incorporated in the MLR model. The VIF test for their multi-collinearity resulted in a range from 1.03 to 2.78 that is well within the acceptable limit 4.0 (Montgomery and Runger 2003). This ensures that all the identified factors can be included in the MLR model.

The MLR analysis results in an equation that describes the statistical relationship between the factors of change (X_n) and the LUCC. The median value of each coefficient set has been utilized in the model. The stability of the mean of the coefficient sets can be inferred from the smaller standard deviations (Table 5) indicating the negligible impact of the sampling performed.

Future city

The weighting parameters calibrated by GA for the MLR identified dominant class conversions viz., vegetation to built-up and other class, water to other, and other to built-up class are 0.5, 20.3, 0.2, and 32.5, respectively. The calibration accuracy (Fig. 4) of the hybrid model was made by comparing the actual and simulated LULC maps (2000 and 2010) of the study area.

Built-up growth and pattern dynamics

Analysis of the cityscape helps to identify the varied urbanization levels and their consequences on the environment. The varied urbanization levels viz., the urban built-up, suburban built-up, and the rural built-up; and the shift in the built-up pattern over time has been illustrated in Fig. 5.

Discussion

SLEUTH—UGM

Clarke (2017) reported that deurbanization and urban growth halting leads to a poor measure of fit at times. Very low

Table 4 Coefficient range and best-fit achieved during the calibration process of the SLEUTH model

Coefficient values	Diffusion	Breed	Spread	Slope resistant	Road gravity	Best-fit Lee–Salle	Best-fit OSM	Computational time (hr)
Coarse calibration						0.350	0.025	6.3736111
Start	0	0	0	0	0			
Step	25	25	25	25	25			
Stop	100	100	100	100	100			
Best-fit	25	75	25	1	75			
Fine calibration						0.320	0.029	5.8330556
Start	25	25	1	1	1			
Step	5	5	10	10	10			
Stop	75	75	75	75	75			
Best-fit	72	72	13	35	50			
Final calibration						0.280	0.029	7.5886111
Start	30	40	13	35	10			
Step	10	10	10	10	10			
Stop	75	75	30	60	70			
Best-fit	1	1	10	30	50			

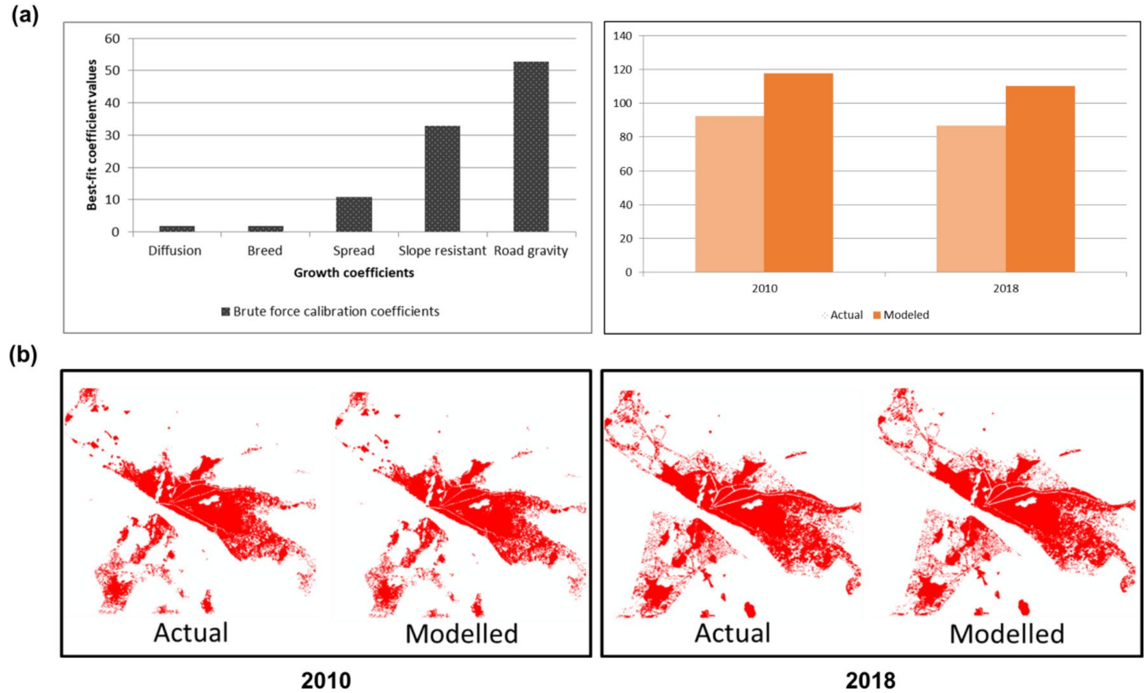

Fig. 2 **a** Best-fit growth coefficients and calibration accuracy achieved with SLEUTH model calibration **b** Validation of the calibrated SLEUTH model

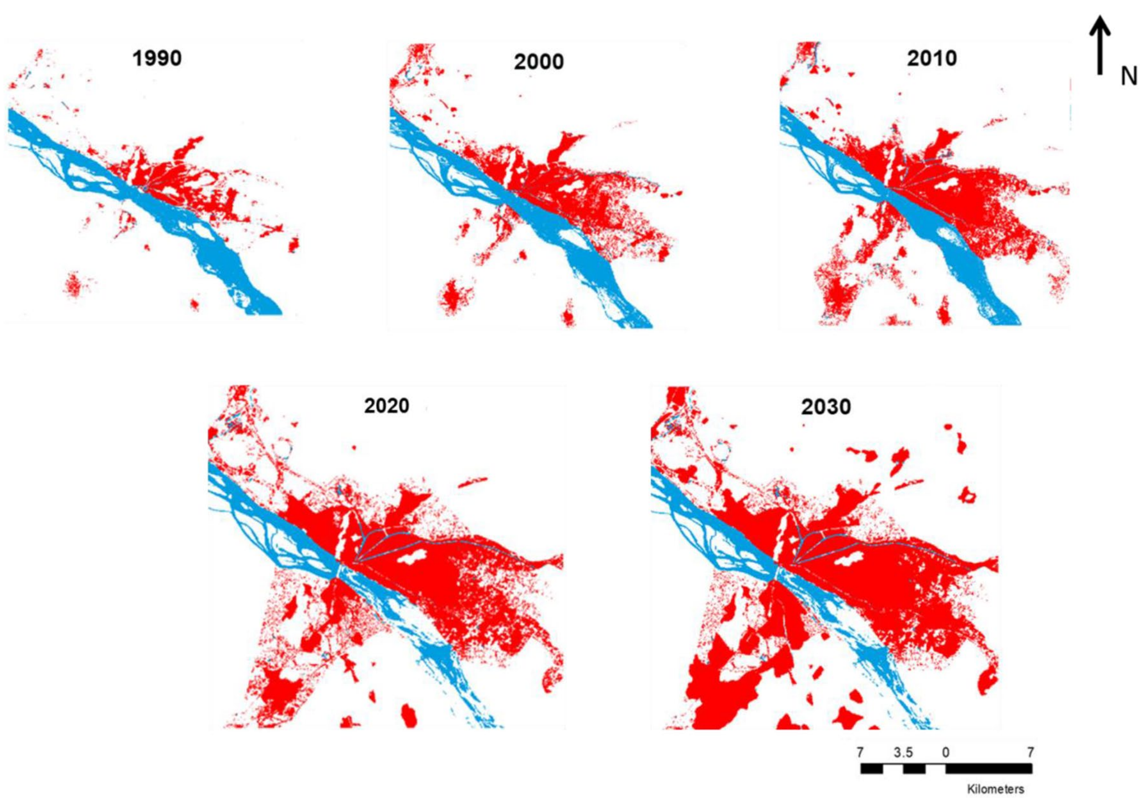

Fig. 3 Urban growth of Vijayawada modelled by the SLEUTH—UGM

Table 5 Coefficient values of the factors of LUCC used in the hybrid model

Factor	Mean coefficient	Standard deviation	Mean p value	Mean standard error
Intercept	0.6989	0.0057	–	0.0071
X_1	1.0514	0.0109	0.0000	0.0127
X_2	− 0.3295	0.0113	0.0000	0.0129
X_3	0.0003	0.0045	0.8214	0.0008
X_4	− 0.0005	0.0012	0.0000	0.0023
X_5	0.0008	0.0019	0.0002	0.0031

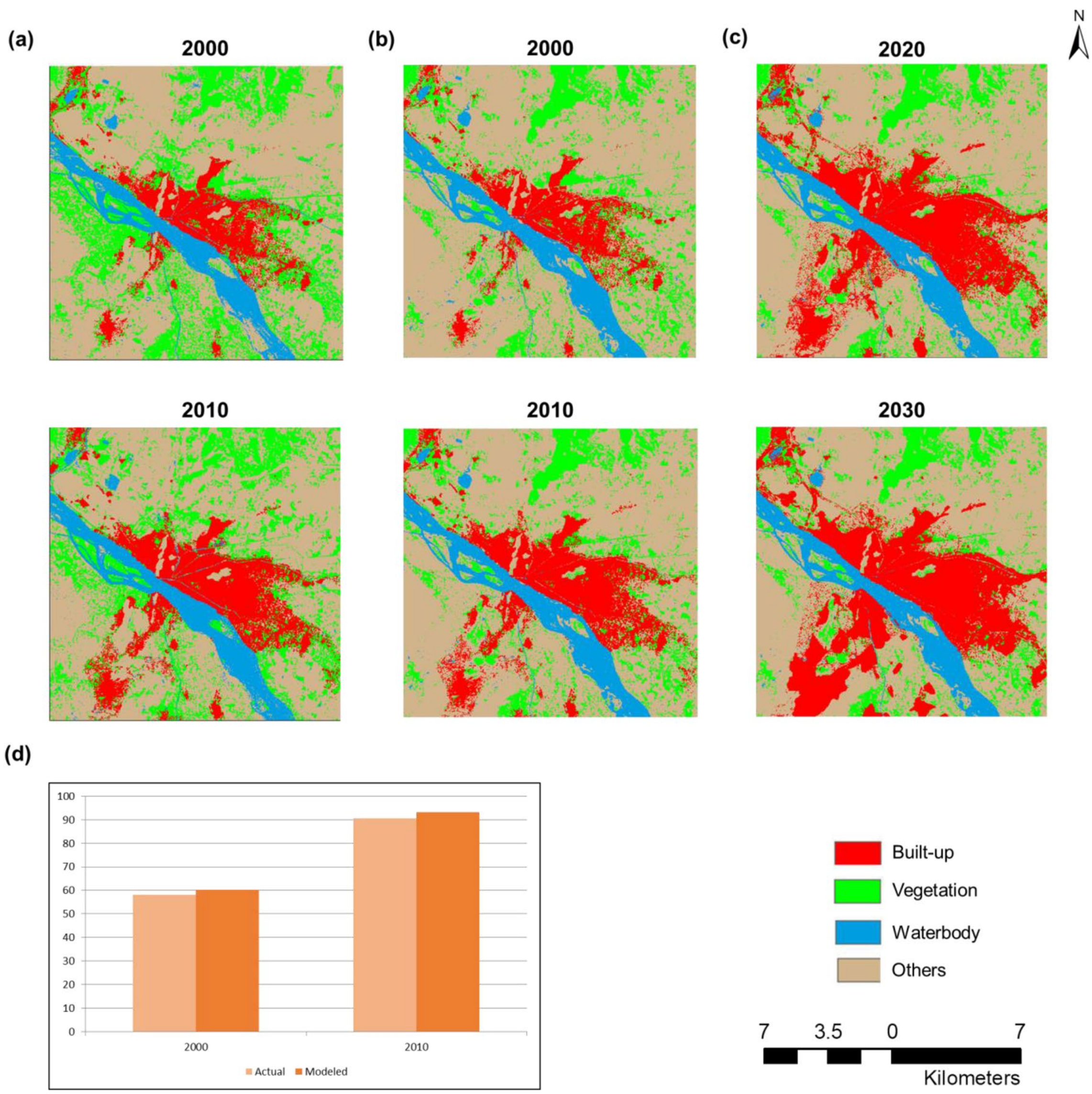

Fig. 4 **a** Actual LULC map derived from Landsat satellite in 2000 and 2010 **b** Simulated LULC map by the hybrid model in 2000 and 2010 **c** LULC scenario developed using the hybrid model for 2020 and 2030 **d** Overall simulation accuracy of the hybrid model captured for the built-up extent

values were observed for the dispersion and breed coefficients (best-fit = 1) at the final calibration phase. This may be related to the lesser spontaneous and new spreading center growths compared to the spreading center and road influenced growth of the city. The modeled urban extent of Vijayawada using SLEUTH during 2020 and 2030 showed that the available open spaces within the existing city extent were converted to built-up, and expansion of built-up class along the fringes of the existing city (Fig. 3). This type of city growth is similar to the historical pattern of the growth of the city matching infill development as observed by Vani and Prasad (2020). Also, the emergence of new urban centers merged with the existing city areas for predictions till 2050 and extended into the dense vegetation on the north resulting in further reduction of green cover around the city. However, only predictions until 2030 were accounted for in this study for comparison with the hybrid model.

The stability of the model to predict the future extent of the city till 2100 was also analysed. It was inferred that the model is more reliable for predicting urban growth of Vijayawada till 2030 and is acceptable with further finer calibrations for predictions only till 2050. A prominent characteristic of the model observed was the model tends to over-estimate built-up areas into clusters with the increasing period of forecasting. Model validation and prediction ensures that SLEUTH is a reliable model to forecast the growth of a city.

Hybrid CA-GA-MLR model

The calculated probability or the p value (Table 5) for each factor tests the null hypothesis that its coefficient equals zero. A lower *p* value of <0.05 indicates that the null hypothesis may be rejected. This explains that a factor with a lower *p* value has a significant contribution to the LUCC process. Conversely, factors with a larger *p* value indicate that the changes in the factors are not associated with the LUCC process and hence can be neglected from including them in the model.

The MLR calibration results indicate class conversion from vegetation and other class to built-up class was observed as dominant for the factors population density, richness index and distance to the road. Slope and population density are the central factors that influence the expansion of built-up areas. Distance to roads is yet another factor that controls the development process as in general, built-up developments are witnessed close to roads. Richness index also exhibits a positive correlation with built-up areas. In contrast, the influence of the city center in accelerating built-up expansion shows an insignificant contribution towards the built-up class. The predictive ability of the MLR model evaluated by pseudo-R-square equals 0.432 that satisfied the good-fit criteria greater than 0.2 as suggested by Clark and Hosking (1986).

Analysis of the transition potential identified a change process dominated by the conversion of vegetation to other class (barren, open and exposed areas, rocks) initially, and then, later into built-up. The LULC predictions with the hybrid model for 2020 and 2030 (Fig. 4) stand effective for built-up expansion with the observed change process that is in concurrence with the study of Vani and Prasad (2020). The predicted output followed a similar pattern wherein developments are more concentrated within the existing built-up areas along with expansion along the existing built-up fringes. Parallel conversion of vegetation class to other class was also dominant in the areas adjoining the built-up areas. It is worth noting that though slope serves as a constraint for development, the expansion of the built-up class on the hilly terrains had been witnessed over the past growth of the city (Fig. 6). This tendency of conversion may adhere to the increasing population and the richness capacity of the individuals.

Cityscape dynamics

The analysis of the built-up dynamics of the study area indicates that the rate of urbanness has been steadily increasing with time. The development pattern witnessed a shift from infill development to expansion in the period 2010–2030. The gradual decrease in the leapfrog development pattern only indicates

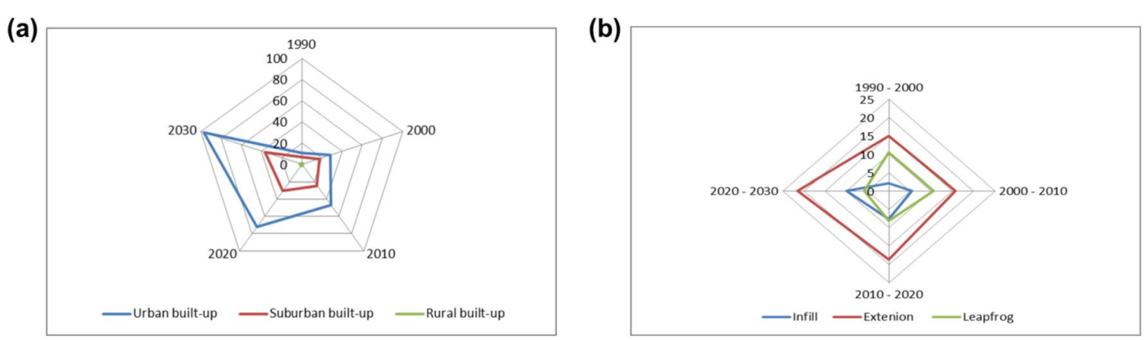

Fig. 5 LUCC dynamics captured in terms of **a** built-up changes **b** development pattern

a more compact development of the city which in turn symbolizes reduction of the open spaces within the city extent.

SLEUTH versus the hybrid model

As this study aims at identifying the dominant driving factors of urban expansion of Vijayawada city, comparison between the SLEUTH and the hybrid model has been carried out. SLEUTH despite being a robust CA-based urban growth model, is not open for incorporating additional driving factors in the model. The driving forces of the SLEUTH model are restricted only to the bio–physical variables of the terrain and do not consider the socio-economic factors that accelerate the growth. This closed nature of the model paved the way to the necessity of a hybrid model that can incorporate socio–economic and allied factors in addition to the bio-physical variables.

The hybrid model used in this study utilizes MLR to automate the calibration of the factors of change with the CA simulating the neighbourhood interactions along with the GA to calibrate the neighbourhood interaction parameters. The hybrid model apart from overcoming the drawbacks of the SLEUTH model, it is open to modifying the drivers of change for LUCC analysis.

Conclusions

Urban sprawl and expansion are global concerns in the process of urbanization. Undoubtedly, studies and researches in these areas will be beneficial to the community involved in planning, policymaking, and managing the development of urban areas. Identification and determination of the significant drivers of LUCC to be included in the model are of great importance and challenge. The SLEUTH model, despite being used widely for modeling urban growth globally, lacks in enabling the modelers to alter the driving forces of change as the model is based on the assumption that the drivers of change are uniform in different locations. It was also observed that the calibration phase is computationally inefficient and requires a large number of CPU hours (Dietzel and Clarke, 2007). As of the study area—Vijayawada—is concerned, though the location comprises many hills, it is least prone to landslides. Hence, the application of the bio-physical drivers'-based SLEUTH to model the city growth is acceptable. It is worth noting that regular encroachment of hills has the potential to alter the natural terrain. However, the model could build scenarios of the future built-up extent without taking into consideration the status of other LULC classes in the respective periods. The CA-based hybrid model developed in combination with the MLR and GA comes handy to overcome the drawbacks of the SLEUTH model. Besides, the model effectively helps in understanding the statistical significance of each factor thus enabling the modeler to eliminate redundant factors that may not accelerate changes in a given area. The modeled scenario for Vijayawada city visualizes the urban planners and policymakers about the possible evolution of the city and aids in planning optimal measures to ensure the sustainability of the available open spaces in and around the city.

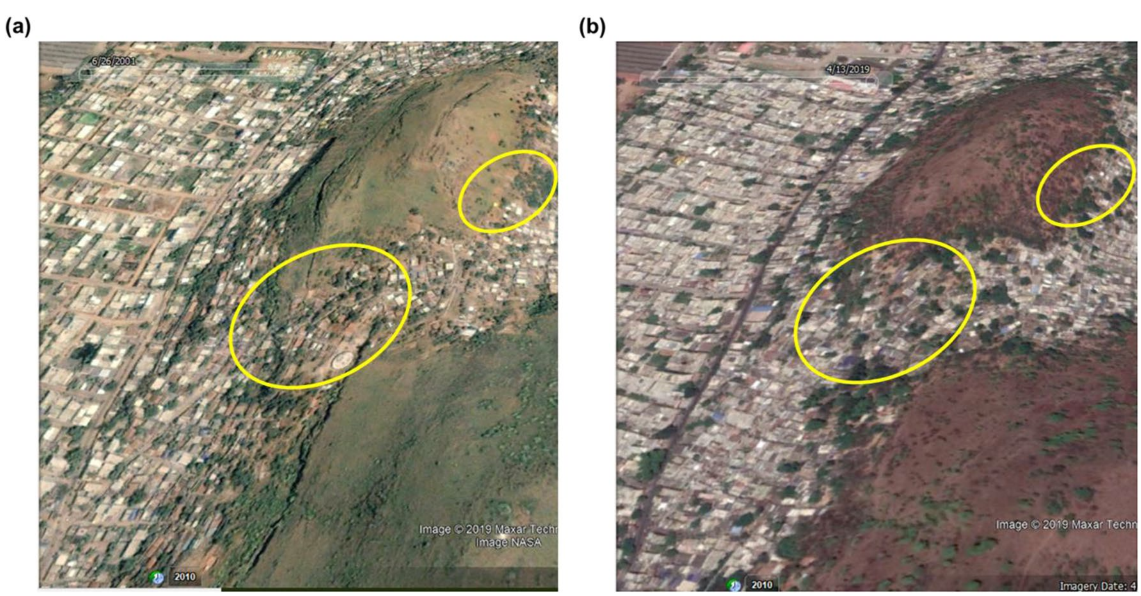

Fig. 6 Built-up expansion along the hill slopes witnessed through Google Earth in **a** 2001 and **b** 2019 Courtesy: Google Earth

Acknowledgements The research was funded by the Human Resource Development Group (HRDG)— Council of Scientific and Industrial Research (CSIR), GoI under Grant [09-917(0011)2K18EMR-I]. The authors thank the United States Geological Survey (USGS) and Google Earth for providing satellite data used in this study. The authors appreciate the Vijayawada Municipal Corporation (VMC) for their inputs and suggestions. The authors also extend their gratitude to Project Gigalopolis for enabling access to the SLEUTH urban growth model. The authors are thankful to Kamakshi Moparthi, Mahindra Ecole University, Hyderabad for her contribution towards the SLEUTH part of the research.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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